



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
REVIEW TEST
20-09-2020

JEE(Main + Advanced) : NURTURE COURSE (PHASE : TEPs)

PAPER-1

Time : 3 Hours

Maximum Marks : 264

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **20** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16,
Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24,
Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127,
Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

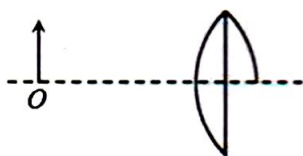
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 40)

- This section contains **TEN** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.

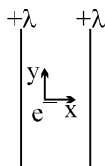
1. A biconvex thin lens is prepared from glass of refractive index $\mu_2 = 3/2$. The two curved surface have equal radii of 20 cm each. One of the surface is silvered from outside to make it reflecting. It is placed in a medium of refractive index $\mu_1 = 5/3$. It acts as a
(A) converging mirror (B) diverging mirror
(C) concave mirror of focal length 12.5 cm (D) convex mirror of focal length 25 cm
2. One fourth part of an equiconvex lens of focal length 100 cm is removed as shown in the figure. An object of height 1cm is placed in front of the lens. It is observed that all the images are of equal height. Then



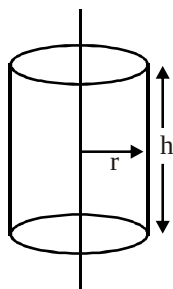
- (A) Object is at a distance of $\frac{400}{3} \text{ cm}$ from the lens.
- (B) The magnitude of magnification produced by upper and lower part is equal.
- (C) The no. of images formed are two
- (D) The product of magnification of both the lenses is negative.

Space for Rough Work

3. An electron is placed just in the middle between two long fixed line charges of charge density $+\lambda$ each. The wires are in the xy plane (Do not consider gravity)



- (A) The equilibrium of the electron will be unstable along x -direction
 (B) The equilibrium of the electron will be neutral along y -direction
 (C) The equilibrium of the electron will be stable along z -direction
 (D) The equilibrium of the electron will be stable along y -direction
4. Consider a finite wire having a uniform charge distribution over its length. It is symmetrically enclosed by coaxial cylindrical Gaussian surface as shown in figure. We apply Gauss's law to find electric field and for that purpose, we wish to evaluate $\oint \vec{E} \cdot d\vec{s}$. Which of the following is the reason for our inability to use Gauss' law.

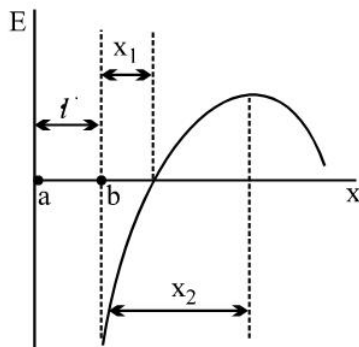


- (A) The flux will be due to the components of electric field normal to the surface only.
 (B) Electric field is varying in magnitude over the surface
 (C) The angle between electric field and area vector is not constant at all the points on the gaussian surface
 (D) The flux through the circular faces will be non zero

Space for Rough Work

5. Two point charges are placed at point a and b. The field strength to the right of the charge Q_b on the line that passes through the two charges varies according to a law that is represented graphically in the figure. Choose the **CORRECT** statement(s):

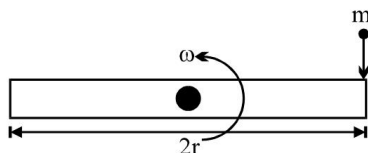
(given: $x_1 = 1\text{m}$, $l = 7\text{m}$)



- (A) Charge on 'a' (Q_a) is positive in nature.
 (B) Charge on 'b' (Q_b) is negative in nature.
 (C) Ratio of magnitudes of charges Q_a/Q_b is 64.
 (D) The distance x_2 is $7/3$ m.
6. A thin spherical glass lens produces a real three times laterally magnified image of an object, when the whole system is in air. If the system, with the same distance between the object and the lens, is immersed in water. ($\mu_{\text{glass}} = 3/2$ and $\mu_{\text{water}} = 4/3$)
- (A) the focal length of the lens increases to 4 times its original value.
 (B) the power of the lens increases to 4 times its original value
 (C) a real magnified image of the object will be formed.
 (D) a virtual magnified image of the object will be formed.

Space for Rough Work

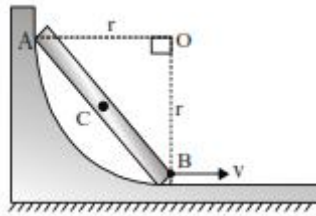
7. A uniform rod of length $2r$ and mass m is rotating in a horizontal plane about a fixed pivot through centre at a steady speed of ω rad/sec. A particle of mass m moving with speed $\omega r/4$ strikes an end of the rod perpendicularly. The rod and the particle are moving towards each other and the coefficient of restitution is $1/2$.



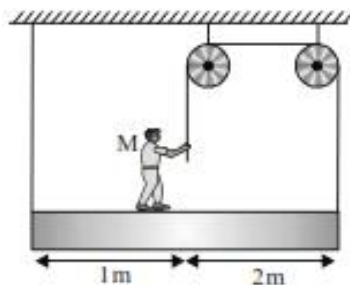
- (A) The impulsive reaction at the pivot is $\frac{15m\omega r}{32} \text{Ns}$.
- (B) The impulsive reaction at the pivot is $\frac{5m\omega r}{32} \text{Ns}$.
- (C) The new angular speed of the rod is $\frac{13\omega}{32} \text{rad/s}$.
- (D) The new speed of particle is $\frac{7r\omega}{32}$.

Space for Rough Work

8. A rigid rod of mass M slides along a fixed semicircular track (in vertical plane) followed by a flat track. At the given instant velocity of end B is v along horizontal plane. Then at the given instant.



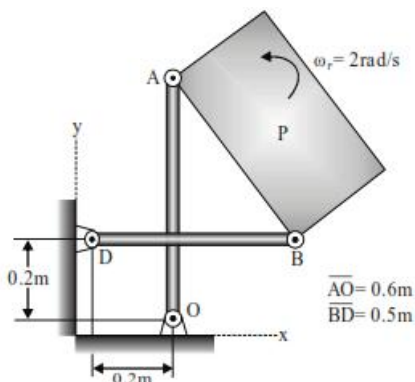
- (A) Angular speed of rod is v/r .
 (B) Velocity of centre of mass is $v/\sqrt{2}$.
 (C) Angular momentum of rod about O is $\frac{2}{3} mvr$.
 (D) The ratio of rotational to translational kinetic energy is 1:6.
9. A system consisting of man on platform is in equilibrium. Mass of man (M) & platform are equal. The platform is in equilibrium due to two massless strings, one attached to fixed support while other string is pulled by a man resting on platform.



- (A) Tension in the left most string is $3Mg/8$.
 (B) Tension in the right most string is $5Mg/8$.
 (C) Normal force between man and platform is $3Mg/8$.
 (D) Tension in the left most string is $3Mg/4$

Space for Rough Work

10. Motion of the rectangular plate P is constrained by two rods OA and BD which cross without touching. For the instant represented where the rods are perpendicular to each other, the plate has a counterclockwise angular velocity $\omega_P = 2 \text{ rad/s}$. At this given instant, choose the **CORRECT** statements: (at all the joint rod and plate can rotate freely)



- (A) Angular velocity of rod OA is $4/3 \text{ rad/sec}$.
 (B) Angular velocity of rod BD is $6/5 \text{ rad/sec}$.
 (C) Velocity of Point 'A' is $4/5 \text{ m/sec}$ in +ve x direction.
 (D) Velocity of Point 'B' is $3/5 \text{ m/sec}$ in +ve y direction.

Space for Rough Work

SECTION-II : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

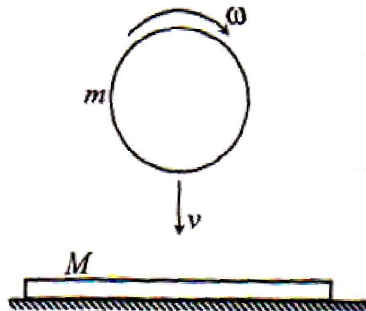
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+											
●	●	0	0	0	0	●	●	●	●	●	●
1	1	1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	●	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9	9	9

-											
●	●	0	0	0	0	●	●	●	●	●	●
1	1	1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	●	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9	9	9

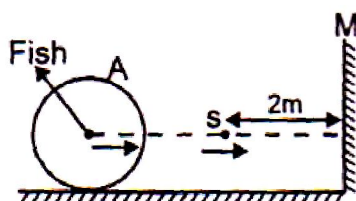
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. A solid ball of mass m and radius r spinning with angular velocity ω falls on horizontal slab of mass M with rough upper surface (coefficient of friction μ) and smooth lower surface. Immediately after collision normal component of velocity of the ball remains half of its value just before collision and it stops spinning. Find the velocity (in S.I. unit) of the sphere in horizontal direction immediately after impact (given: $R\omega = 5$ in S.I. unit)



Space for Rough Work

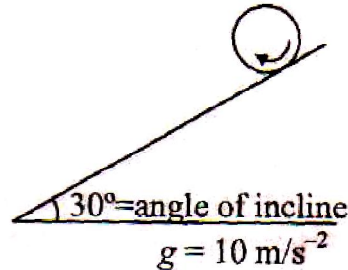
2. Find the electric flux (in Nm^2/C) passing through an infinite strip of small width h at a distance d ($d \gg h$) from a point charge q . Take $q = 1\mu\text{C}$, $h = 1\text{cm}$, $d = 1\text{m}$.
3. A is a thin walled sphere at rest made up of glass. The radius of the sphere is 1m and it is filled with a transparent liquid of refractive index μ . S is the luminous source moving directly towards a plane vertical mirror M. A fish in the liquid is moving towards S. The eye of the fish and S are collinear & perpendicular to mirror M. At the moment S is 3m away from the centre of the sphere fish observes that image of S due to reflection is moving with speed of 13m/s . If speed of the fish relative to the sphere is 10 m/s and $\mu = 1.5$ then find the speed (in m/s) of the source at that instant. The system is placed in air.



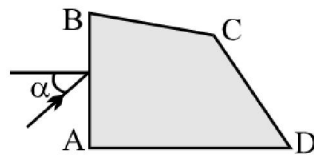
4. A fixed circle of radius a and centre O is drawn and a charge $+Q_1$ is placed at a distance $3a/4$ from O on a line through O perpendicular to the plane of circle. If a second charge Q_2 is similarly placed at a distance $\frac{5a}{12}$ on the opposite side of circle, the net electric flux through circle becomes zero. Find value of $\frac{Q_1}{Q_2}$?

Space for Rough Work

5. A homogenous solid cylinder of radius 2m is spinned to an angular speed 10 rad s^{-1} and then placed on an inclined plane for which coefficient of friction is $\mu = \tan 30^\circ$. Calculate the time (in sec.) for which the cylinder will be unmoved.

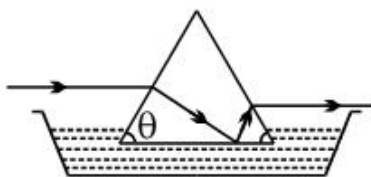


6. The faces of prism ABCD made of glass of refractive index ($\mu = \sqrt{3}$) from dihedral angles $\angle A = 90^\circ$, $\angle B = 75^\circ$, $\angle C = 135^\circ$ and $\angle D = 60^\circ$. A beam of light falls on face AB and after complete internal reflection from face BC, escapes through face AD. Find value of $\sin(\alpha)$, if a beam that has passed through the prism in this manner, is perpendicular to the incident beam.



Space for Rough Work

7. A plane mirror 50 cm long, is hung parallel to a vertical wall of a room, with its lower edge 50cm above the ground. A man stands in front of the mirror at a distance 2 m away from the mirror. If his eyes are at a height 1.8 m above the ground, find the length (in m) of the floor between him & the mirror, visible to him reflected from the mirror.
8. An isosceles triangular glass prism stands with its base in water as shown. The angles that its two equal sides make with the base are θ each. An incident ray of light parallel to the water surface internally reflects at the glass-water interface and subsequently re-emerges into the air. Taking the refractive indices of glass and water to be $3/2$ and $4/3$ respectively, find maximum possible value of $\cos^2 \theta$ for this to happen?



Space for Rough Work

SECTION-III : Integer Value Correct Type
No question will be asked in section III

SECTION-IV : (Maximum Marks : 16)

- This section contains **TWO** questions.
- Each question contains two columns, **Column-I** and **Column-II**.
- **Column-I** has **four** entries (A), (B), (C) and (D)
- **Column-II** has **five** entries (P), (Q), (R), (S) and (T)
- Match the entries in **Column-I** with the entries in **column-II**.
- One or more entries in **Column-I** may match with one or more entries in **Column-II**.
- The ORS contains a 4×5 matrix whose layout will be similar to the one shown below :

(A)	☐ (P)	☐ (Q)	☐ (R)	☐ (S)	☐ (T)
(B)	☐ (P)	☐ (Q)	☐ (R)	☐ (S)	☐ (T)
(C)	☐ (P)	☐ (Q)	☐ (R)	☐ (S)	☐ (T)
(D)	☐ (P)	☐ (Q)	☐ (R)	☐ (S)	☐ (T)

- For each entry in **column-I**, darken the bubbles of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q), (R) and (T), then darken these three bubbles in the ORS. Similarly, for entries (B), (C) and (D).
- For each question, marks will be awarded in one of the following categories :

For each entry in **Column-I**

Full Marks : +2 If only the bubble(s) corresponding to all the correct match(es) is (are) darkened

Zero Marks : 0 If none of the bubbles is darkened

Negative Marks : -1 In all other cases

1. A positive point charge is given an initial velocity $3\hat{i} + 4\hat{j}$ m/s in a uniform electric field $E = -100\hat{i} + 50\hat{j}$ N/C. Initial position of the charge is in 4th quadrant.

Column-I	Column-II
(A) speed of particle	(P) Remains constant
(B) Y component of velocity	(Q) Never zero
(C) x-coordinate of the particle	(R) Always +ve
(D) Magnitude of acceleration	(S) First increases then decreases
	(T) First decreases then increases

Space for Rough Work

2. A prism of refracting angle A is made of material of refractive index 2. Assume rays are incident at angle of incidence i such that $0 \leq i \leq 90^\circ$.

Column-I

- (A) $A = 15^\circ$
 (B) $A = 45^\circ$
 (C) $A = 70^\circ$
 (D) $A = 50^\circ$

Column-II

- (P) All rays are reflected back from second surface.
 (Q) All rays are refracted into air from second surface.
 (R) Some rays are reflected back from second surface.
 (S) Some rays are refracted into air from second surface
 (T) No total internal reflection takes place at the second surface.

Space for Rough Work

PART-2 : CHEMISTRY**SECTION-I : (Maximum Marks: 40)**

- This section contains **TEN** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.

-
1. Which pair of species contain same type of anion.
(A) NaCl and PCl_5 (solid)
(B) NaNO_3 and N_2O_4 (solid)
(C) KBr and PBr_5 (solid)
(D) KClO_4 and Cl_2O_6 (solid)
 2. Select **CORRECT** name of salt.
(A) Potassium dihydrogenhypophosphate (B) Sodium dihydrogenpyrophosphite
(C) Sodium hydrogenperoxydisulphate (D) Potassium dihydrogenpyrophosphate
 3. Which of the following is/are **INCORRECT** regarding real gas ? (T_B = Boyle temperature)
(A) If $\lim_{P \rightarrow 0} \left(\frac{\partial Z}{\partial P} \right)_T < 0$ for a particular gas then $T < T_B$ & repulsive part of potential dominates.
(B) If $\lim_{P \rightarrow 0} \left(\frac{\partial Z}{\partial P} \right)_T < 0$ for a particular gas then $T > T_B$ & attractive part of potential dominates.
(C) If $\lim_{P \rightarrow 0} \left(\frac{\partial Z}{\partial P} \right)_T > 0$ for a particular gas then $T < T_B$ & attractive part of potential dominates.
(D) If $\lim_{P \rightarrow 0} \left(\frac{\partial Z}{\partial P} \right)_T > 0$ for a particular gas then $T > T_B$ & repulsive part of potential dominates.

Space for Rough Work

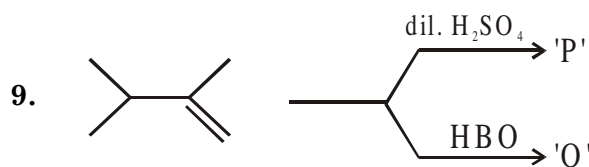
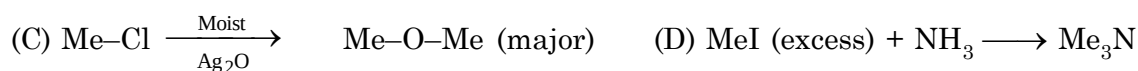
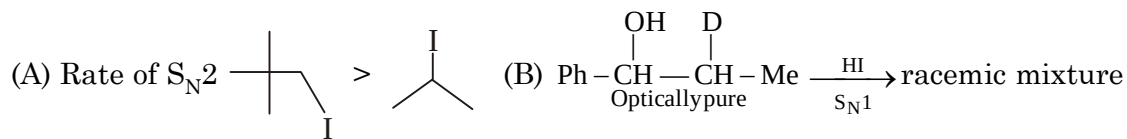
4. Which of the following statements are **FALSE**
- (A) The electron probability density at the nucleus is zero for all H-atom states
 - (B) In the ground state of an H-atom, the electron is restricted to move on the surface of sphere
 - (C) For the ground state H-atom, the electron probability density is greater than zero at all location in the atom
 - (D) All the spherical harmonics (Angular part of wave function) are constant on the surface of a sphere centered at the origin.
5. Positive deviation from ideal behaviour take place because of
- (A) Molecular interaction between atom and $\frac{PV}{nRT} > 1$
 - (B) Finite size of atom and $\frac{PV}{nRT} > 1$
 - (C) $(b - a/RT) > 0$ & $\frac{PV}{nRT} < 1$
 - (D) $(b - a/RT) > 0$ & $\frac{PV}{nRT} > 1$

Space for Rough Work

6. For dissociation of a gas $\text{N}_2\text{O}_5(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g})$. The reaction is performed at constant temperature and volume. If D is the vapour density of equilibrium mixture, P_0 is initial pressure of $\text{N}_2\text{O}_5(\text{g})$ and M is molecular mass of N_2O_5 , then the correct information(s) at the equilibrium is/are
- (A) The total pressure of gases at equilibrium is $\frac{P_0 \cdot M}{2D}$.
- (B) The degree of dissociation of $\text{N}_2\text{O}_5(\text{g})$ is $\frac{M - 2D}{3D}$.
- (C) The partial pressure of $\text{N}_2\text{O}_5(\text{g})$ at equilibrium is $\frac{(5D - M) \cdot P_0}{3D}$.
- (D) The partial pressure of $\text{O}_2(\text{g})$ at equilibrium is $\frac{(M - 2D) \cdot P_0}{3D}$.
7. Which reaction not giving $\text{S}_{\text{N}}2$ product as a major product ?
- (A) 1-bromobutane reacts with sodium methoxide
- (B) 2-bromobutane reacts with large excess of potassium tertbutoxide
- (C) 2-bromo 1, 1-dimethyl cyclopentane reacts with ethanol
- (D) Bromocyclohexane reacts in methanol at high temperature

Space for Rough Work

8. Choose the incorrect option(s).



Correct statement regarding P & Q is/are

- (A) P & Q are structural isomer (B) P can show optical isomerism
 (C) Q can show optical isomerism (D) Q is 3° alcohol

10. Which of the following option is **INCORRECT** :

- (A) To prepare t-butyl ethyl ether the appropriate reagents will be $\text{C}_2\text{H}_5\text{ONa}$ & $\text{CH}_3-\overset{\text{CH}_3}{\underset{\text{CH}_3}{\text{C}}}-\text{Cl}$
 (B) Cyclohexanone forms cyanohydrin in good yield but 2, 2, 6-trimethyl cyclohexanone does not
 (C) CHCl_3 on aerial oxidation poisonous gas CO is formed
 (D) To prepare t-butyl bromide from isobutyl bromide the reagents used will be alcoholic KOH followed by Br_2

Space for Rough Work

SECTION-II : (Maximum Marks: 32)

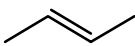
- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

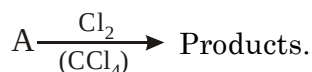
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

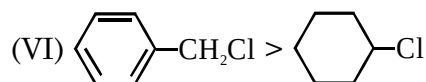
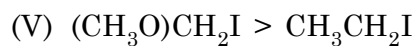
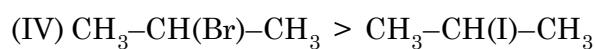
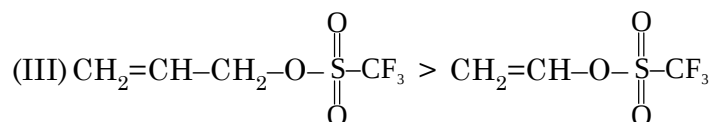
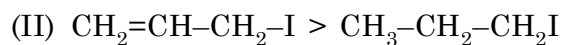
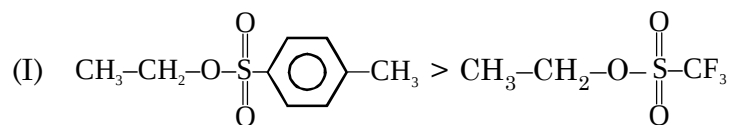
- Total number of alcohol which will give  as major product with conc. H₂SO₄
- Number of optically active isomers of formula C₇H₁₆ will be ?
- 1, 3-butadiene $\xrightarrow[\text{(CHCl}_3\text{)}]{\text{Cl}_2}$ A(1, 2-addition product) + B & C (1, 4 addition product)



What is the number of products.

Space for Rough Work

4. How many given orders is/are correct for rate of nucleophilic substitution reaction under identical conditions:



Space for Rough Work

5. The quantum number ' n ' corresponds to the excited state of He^+ ion if on transition to the ground state that ion emits two photons in succession with wavelength 108.5 nm and 30.4 nm. The magnitude of energy of electron (eV) corresponding to this state is.
6. A cylindrical container with a moveable piston initially holds 1.5 mol of gas at a pressure of 4.0 atm and a volume of 2.5 L. If the piston is moved to final volume of 5.0 L, while simultaneously withdrawing 0.75 moles of gas, what is the final pressure (in atm) ?
7. Find maximum number of 'S' atoms that must lie in same plane in $\text{S}_4\text{O}_6^{2-}$ oxyanion.
8. How many total unshared oxygen atoms are in Benitoite ($\text{BaTiSi}_3\text{O}_9$) ?

Space for Rough Work

SECTION-III : Integer Value Correct Type
No question will be asked in section III

SECTION-IV : (Maximum Marks : 16)

- This section contains **TWO** questions.
- Each question contains two columns, **Column-I** and **Column-II**.
- **Column-I** has **four** entries (A), (B), (C) and (D)
- **Column-II** has **five** entries (P), (Q), (R), (S) and (T)
- Match the entries in **Column-I** with the entries in **column-II**.
- One or more entries in **Column-I** may match with one or more entries in **Column-II**.
- The ORS contains a 4×5 matrix whose layout will be similar to the one shown below :

(A)	☐ (P)	☐ (Q)	☐ (R)	☐ (S)	☐ (T)
(B)	☐ (P)	☐ (Q)	☐ (R)	☐ (S)	☐ (T)
(C)	☐ (P)	☐ (Q)	☐ (R)	☐ (S)	☐ (T)
(D)	☐ (P)	☐ (Q)	☐ (R)	☐ (S)	☐ (T)

- For each entry in **column-I**, darken the bubbles of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q), (R) and (T), then darken these three bubbles in the ORS. Similarly, for entries (B), (C) and (D).
- For each question, marks will be awarded in one of the following categories :

For each entry in **Column-I**

Full Marks : +2 If only the bubble(s) corresponding to all the correct match(es) is (are) darkened

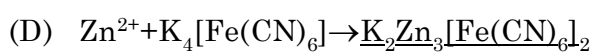
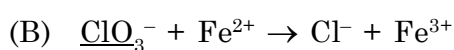
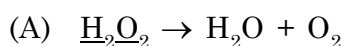
Zero Marks : 0 If none of the bubbles is darkened

Negative Marks : -1 In all other cases

1. Match the column :

Column-I

(Chemical reactions)
given chemical



Column-II

('n' factor of underlined specie in the redox reactions)

(P) 4

(Q) 1

(R) 2

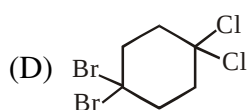
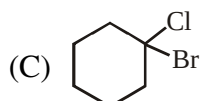
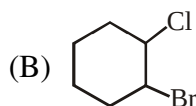
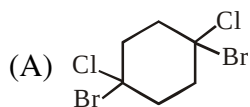
(S) 3

(T) 6

Space for Rough Work

2. Match the Column :

Column - I



Column - II

(P) Bromochloro cyclohexane & can show stereoisomerism

(Q) Dibromo dichloro cyclohexane & show stereo isomerism

(R) Optically active

(S) Substituted cyclohexane showing G.I.

(T) Substituted cyclohexane not showing stereoisomerism

Space for Rough Work

PART-3 : MATHEMATICS**SECTION-I : (Maximum Marks: 40)**

- This section contains **TEN** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.

-
1. Let $y^2 - 5y + 3x + k = 0$ be a parabola, then -
(A) Its latus rectum is least when $k = 1$
(B) Its latus rectum is independent of k
(C) The line $y = 2x + 1$ will touch parabola if $k = \frac{73}{16}$
(D) $y = \frac{5}{2}$ is the only normal to parabola whose slope is zero
2. Maximum length of chord of the ellipse $\frac{x^2}{8} + \frac{y^2}{4} = 1$, such that eccentric angles of its extremities differ by $\frac{\pi}{2}$ is -
(A) 4 (B) $2\sqrt{2}$ (C) 16 (D) 8
-

Space for Rough Work

3. Two tangents PA and PB are drawn to circle $S = 0$ where $P(4,0)$. If line joining to point P and centre Q cuts the circle at $C(2,0)$ and line AB at $\left(\frac{4}{3}, 0\right)$ then -
- (A) radius of circle is 1 unit (B) radius of circle is 2 units
- (C) $\frac{\text{area of } \Delta PAB}{\text{area of } \Delta AQB} = 8$ (D) $\frac{\text{area of } \Delta PAB}{\text{area of } \Delta AQB} = 4$
4. If r_1, r_2, r_3 be the radii of three circles $S_1 = 0$, $S_2 = 0$, $S_3 = 0$ with their centres $A(2,2)$, $B(4,0)$, $C(1,-3)$ respectively, such that $r_1 : r_2 : r_3 \equiv 1 : 1 : 2$ and $S_2 = 0$, $S_3 = 0$ touches each other externally, then equation of line passing through the common point of $S_2 = 0$ and $S_3 = 0$ and touching the circle $S_1 = 0$, is-
- (A) $x + y - 1 = 0$ (B) $2x - y - 2 = 0$ (C) $x + y - 2 = 0$ (D) $7x - y - 22 = 0$
5. Let n denotes number of four digits numbers using 0,1,2,3,4,5 (repetition of digits is not allowed)-
- (A) $n = 120$
- (B) If product of digits is divisible by 5, then $n = 276$
- (C) $n = 300$
- (D) If product of digits is divisible by 5, then $n = 204$
6. Sum of the series $({}^{100}C_1)^2 + 2.({}^{100}C_2)^2 + 3.({}^{100}C_3)^2 + \dots + 100.({}^{100}C_{100})^2$ equals-
- (A) $\frac{2^{99} [1.3.5 \dots (199)]}{99!}$ (B) $100. {}^{200}C_{100}$
- (C) $50. {}^{200}C_{100}$ (D) $100. {}^{199}C_{99}$

Space for Rough Work

7. Let $x = a\pi$ be the solution of the equation $\sec x + \operatorname{cosec} x = \sin x + \cos x$ where $x \in [0, 2\pi]$, then identify the correct statement(s) -
- (A) a must be greater than $1/2$ (B) a must be less than 1
 (C) a must be greater than 1 (D) a can not be an integer
8. The system of equations $x - y + 3z = 2$, $2x - y + z = 4$ & $x - 2y + \alpha z = 3$ has -
- (A) unique solution only for $\alpha = 0$ (B) unique solution for $\alpha \neq 8$
 (C) infinite solutions for $\alpha = 8$ (D) No solution for $\alpha = 8$
9. If a, b, c are in A.P., b, c, d are in G.P. & c, d, e are in H.P., then -
- (A) a, e, c are in G.P. (B) a, c, e are in G.P.
 (C) $\frac{c+a}{c-a} = \frac{e+c}{e-c}$ (D) $\frac{e+c}{e-c} = \frac{a+e}{a-e}$
10. Words are formed by using all the letters of the word AAIIOCEAL. Which of the following statement(s) is/are correct ?
- (A) Number of words which contain the word 'IOCL' is $\frac{5!}{3!}$
 (B) Number of words in which the word 'IOCL' is NOT at last four places is $\frac{4!}{3!}$
 (C) Number of words in which first three letters are NOT 'A' is ${}^5C_3 \cdot 5!$
 (D) Number of words in which letters of the word 'IOCL' are separated is $5 \cdot \frac{4!}{3!}$.

Space for Rough Work

SECTION-II : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	●	●	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	●	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. If the area of a triangle is 4 sq. units and one of its angle is $\frac{\pi}{6}$ such that the side opposite to the given angle is minimum then the sum of the lengths of the other sides is equal to
2. In an equilateral ΔABC , D is a point on BC such that ratio of inradius of ΔABD to inradius of ΔADC is 2 : 1. Minimum value of $\frac{BD}{CD} + \frac{AD}{BD}$ is
3. Few lines are drawn in a plane such that no two are parallel. Another k lines are drawn parallel to each other in same plane but not parallel to the original lines, there are 8 lines in total and no three are concurrent and maximum 40 distinct triangles can be formed then 'k' is

Space for Rough Work

4. If system of equation
$$\begin{aligned} [a]x + [a-1]y &= 1 \\ x + y &= 1 \\ [a-1]x - [a-2]y &= 1 \end{aligned}$$
 is consistent (where $[.]$ denotes greatest integer function) then number of integral possible values of 'a' is
5. If the solution of $\sin^2 x + \cot^2 x = 3$, where $x \in \left(0, \frac{\pi}{2}\right)$ is $x = k$, then $\operatorname{cosec}^2 k$ is equal to
6. If $f_n(\theta) = \sum_{r=1}^n \frac{\cos(2r\theta)}{\sin(2r+1)\theta \sin(2r-1)\theta}$ then $f_{29}\left(\frac{\pi}{6}\right)$ is
7. Let ordered pair (x, y) satisfies the equation $x^{\log_{10} y} = 2$ and $xy = 20$, then the value of $100\left(\frac{\Sigma x}{\Sigma y}\right)$ is equal to
8. Sum of all integral value(s) of k for which $|x-1| + |x^2-4| + |x+1| = k$ has 2 distinct real solutions in the interval $x \in [-3, 4]$, is

Space for Rough Work

SECTION-III : Integer Value Correct Type
No question will be asked in section III

SECTION-IV : (Maximum Marks : 16)

- This section contains **TWO** questions.
- Each question contains two columns, **Column-I** and **Column-II**.
- **Column-I** has **four** entries (A), (B), (C) and (D)
- **Column-II** has **five** entries (P), (Q), (R), (S) and (T)
- Match the entries in **Column-I** with the entries in **column-II**.
- One or more entries in **Column-I** may match with one or more entries in **Column-II**.
- The ORS contains a 4×5 matrix whose layout will be similar to the one shown below :

(A)	☐ (P)	☐ (Q)	☐ (R)	☐ (S)	☐ (T)
(B)	☐ (P)	☐ (Q)	☐ (R)	☐ (S)	☐ (T)
(C)	☐ (P)	☐ (Q)	☐ (R)	☐ (S)	☐ (T)
(D)	☐ (P)	☐ (Q)	☐ (R)	☐ (S)	☐ (T)

- For each entry in **column-I**, darken the bubbles of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q), (R) and (T), then darken these three bubbles in the ORS. Similarly, for entries (B), (C) and (D).
- For each question, marks will be awarded in one of the following categories :

For each entry in **Column-I**

Full Marks : +2 If only the bubble(s) corresponding to all the correct match(es) is (are) darkened

Zero Marks : 0 If none of the bubbles is darkened

Negative Marks : -1 In all other cases

1.	Column-I	Column-II
(A)	If the image of (3, 8) in the line $x + 3y = 7$ is (a,b), then $a - b$ is equal to	(P) 3
(B)	If the points (ℓ, ℓ) , $(\ell, 2\ell)$ & $(2\ell, \ell)$ enclose a triangle of area 18 sq. units and (a,b) is its centroid then $\frac{ a + b }{4}$ is equal to	(Q) 4
(C)	If the area enclosed between the pair of tangents drawn from (4,5) to the circle $x^2 + y^2 - 2x - 2y - 7 = 0$ & the given circle is equal to $k - 9t$ where $\tan t = \frac{4}{3}$ then $\frac{k}{2}$ is equal to	(R) 6
(D)	Area of the triangle formed by the lines $x^2y + x^2 - y^3 + 3y^2 - 4 = 0$ is equal to	(S) Odd number (T) Prime number

Space for Rough Work

2.	Column-I	Column-II
(A)	Number of geometric progression with 8, 12 & 27 as its three terms is equal to	(P) 0
(B)	In a triangle ABC, $\frac{b^2 \cos 2A - a^2 \cos 2B}{b^2 - a^2}$ ($b \neq a$) is equal to	(Q) 1
(C)	A straight line is drawn from the mid point of one of the equal sides of an right isosceles triangle to the opposite vertex such that it divides the opposite angle into two parts θ_1 & θ_2 where $\theta_1 > \theta_2$, then $\frac{\tan \theta_1}{\tan \theta_2}$ is equal to	(R) $3/2$
(D)	Number of solutions of $x^2 + 9 = 6x \cos x$ in $[0, \pi]$ is equal to	(S) an odd number (T) infinite

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has three sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I	One or more correct option(s)	10	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	No Partial Marking Applicable	0 If none of the bubbles is darkened	-2 In all other cases	40
II	Numerical Value Type (Up to second decimal place)	8	+4 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	32
IV	Matrix Match Type	2	+8 If only the bubble(s) corresponding to all the correct match(es) is(are) darkened	+2 For darkening a bubble corresponding to each correct match is darkened	0 If none of the bubbles is darkened	-1 In all other cases	16

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator